

2D Array Interview Practice Problems (Hints, TC, SC, Time)

Problem 1: Print Matrix Row-Wise

Hint: Use two nested loops. Outer loop for rows, inner loop for columns.

TC: $O(n \times m)$

SC: $O(1)$

Estimated Time: 5 minutes

Problem 2: Sum of Each Row

Hint: For each row, keep a running sum while iterating columns.

TC: $O(n \times m)$

SC: $O(1)$

Estimated Time: 7 minutes

Problem 3: Search an Element in 2D Array

Hint: Traverse row by row and compare each element with target.

TC: $O(n \times m)$

SC: $O(1)$

Estimated Time: 8 minutes

Problem 4: Find Largest Element in Each Column

Hint: Fix a column and iterate through all rows to find max.

TC: $O(n \times m)$

SC: $O(1)$

Estimated Time: 10 minutes

Problem 5: Count Even and Odd Numbers

Hint: Check each element using modulo operator.

TC: $O(n \times m)$

SC: $O(1)$

Estimated Time: 8 minutes

Problem 6: Diagonal Sum of Matrix

Hint: Primary diagonal: $i == j$, Secondary diagonal: $i + j == n - 1$.

TC: $O(n)$

SC: $O(1)$

Estimated Time: 12 minutes

Problem 7: Row with Maximum Sum

Hint: Maintain maxSum and rowIndex while calculating each row sum.

TC: $O(n \times m)$
SC: $O(1)$
Estimated Time: 10 minutes

Problem 8: Transpose of a Matrix

Hint: Swap rows and columns. For new matrix, use $\text{result}[j][i] = \text{original}[i][j]$.
TC: $O(n \times m)$
SC: $O(n \times m)$
Estimated Time: 12 minutes

Problem 9: Spiral Matrix Traversal

Hint: Use four boundaries: top, bottom, left, right and shrink them.
TC: $O(n \times m)$
SC: $O(1)$
Estimated Time: 20 minutes

Problem 10: Search in a Sorted Matrix

Hint: Start from top-right or bottom-left and eliminate rows/columns.
TC: $O(n + m)$
SC: $O(1)$
Estimated Time: 20 minutes